Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
Steps to Solve a Multi-Step Equation	1. Distribute (if necessary) 2. Combine like terms. 3. Solve!
Examples	

1. $9x + 1 - 7x - 5 = -20$

$2x - 4 = -20$

$+4 \quad +4$

$\frac{2x}{2} = \frac{-16}{2}$

$x = -8$

2. $91 = -7(3a - 1)$

$91 = -21a + 7$

$-7 \quad -7$

$\frac{84}{-21} = \frac{-21a}{-21}$

$a = -4$

3. $4m - 5(3m + 10) = 126$

$4m - 15m - 50 = 126$

$-11m - 50 = 126$

$+50 \quad +50$

$\frac{-11m}{-11} = \frac{176}{-11}$

$m = -16$

4. $-3(k - 8) - (k + 5) = 23$

$-3k + 24 - k - 5 = 23$

$-4k + 19 = 23$

$-19 \quad -19$

$\frac{-4k}{-4} = \frac{4}{-4}$

$k = -1$

5. $20 = 10x - 6(2x + 5)$

$20 = 10x - 12x - 30$

$20 = -2x - 30$

$+30 \quad +30$

$\frac{50}{-2} = \frac{-2x}{-2}$

$x = -25$

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6. $8(2w-1)-4w=-116$

$16w-8-4w=-116$

$12w-8=-116$

$+8 \quad +8$

$\frac{12w}{12}=\frac{-108}{12}$

$w=-9$

7. $11h-(2h-1)=118$

$11h-2h+1=118$

$9h+1=118$

$-1 \quad -1$

$\frac{9h}{9}=\frac{117}{9}$

$h=13$

8. $-25=\frac{1}{2}(10x-2)+3x$

$-25=5x-1+3x$

$-25=8x-1$

$+1 \quad +1$

$\frac{-24}{8}=\frac{8x}{8}$

$x=-3$

9. $7-\frac{5}{2}(8r-6)+2r=32$

$7-20r+15+2r=32$

$-18r+22=32$

$-22 \quad -22$

$\frac{-18r}{-18}=\frac{10}{-18}$

$r=-\frac{5}{9}$

Translate & Solve



10. *"Five times the difference of twice a number and three, decreased by the sum of the number and eight, equals 13."*

$5(2n-3)-(n+8)=13$

$10n-15-n-8=13$

$9n-23=13$

$+23 \quad +23$

$\frac{9n}{9}=\frac{36}{9}$

$n=4$

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